Question:

Blake and Kelly are having a contest to see who can run the most in 15 minutes. They decide to do it on a football field that is 100 yards long. Blake runs back and forth 15 times. Kelly runs back and forth once, and then decides that she doesn't want to run next to Blake, so she starts to run to the 40-yard line and back. She does this 34 times. How much farther does the winner run than the loser?

Reasoning + Answer:

<think>To solve this problem, let's first calculate the total distance Blake runs. Since the football field is 100 yards long and Blake runs back and forth 15 times, we can determine the distance he covers by multiplying the number of back-and-forths by the length of the field. So, let's multiply 15 by 100: \(15 \times 100 = 1500 \) yards. This tells us that Blake runs a total of 1500 yards. Next, let's calculate the distance Kelly runs. Kelly runs back and forth once first, so we calculate the distance for this part as well: \(1 \times 100 = 100 \) yards. Then, she runs to the 40-yard line and back to the starting point of her run, which is 80 yards one way. Since she does this 34 times, we need to calculate the total distance for these 34 runs. First, let's find out the total distance for one complete round of her new pattern (40 yards back and forth), which is \(2 \times 40 = 80 \) yards. Now, we multiply this by 34 to find the total distance of her 34 runs: \(80 \times 34 = 2720 \) yards. Now, to find the total distance Kelly runs, we add the initial run to the distance of her 34 runs: \(100 + 2720 = 2820 \) yards. Finally, to determine how much farther the winner ran compared to the loser, we need to compare the totals of Blake and Kelly. We have calculated that Blake ran 1500 yards and Kelly ran 2820 yards. To find out how much farther Kelly ran, we subtract Blake's total distance from Kelly's total distance: \(2820 - 1500 = 1320 \) yards.</think><answer> 1320</answer>